

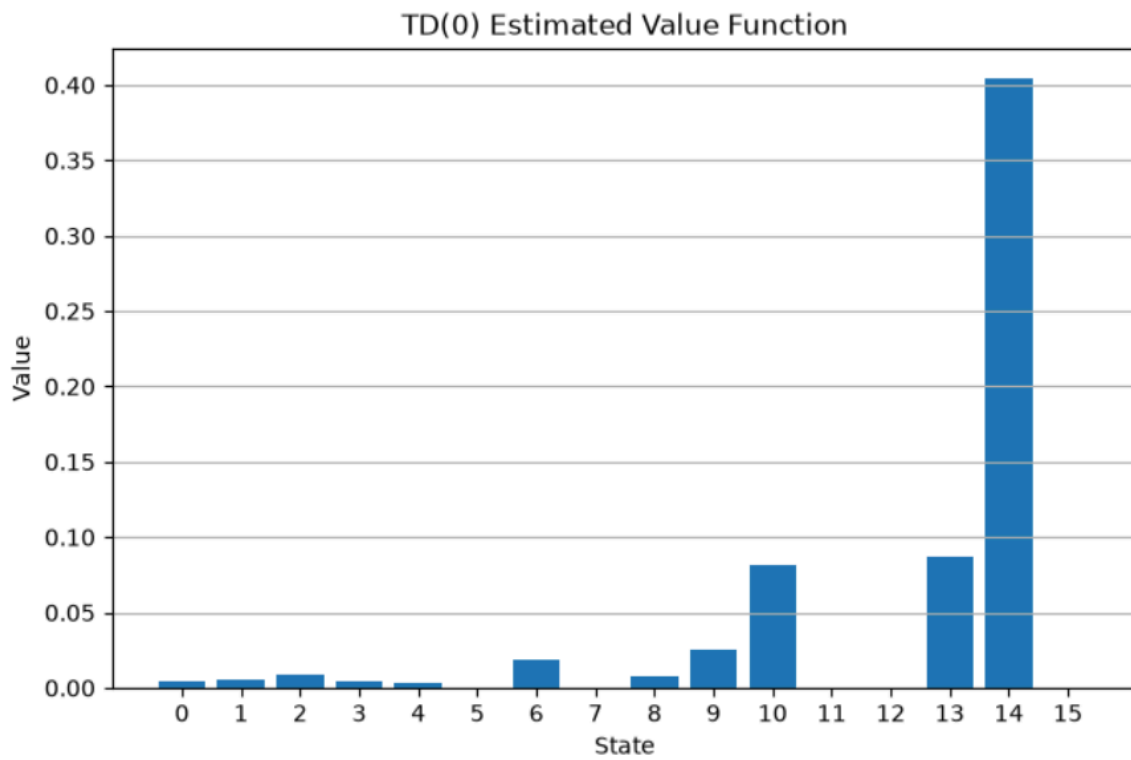
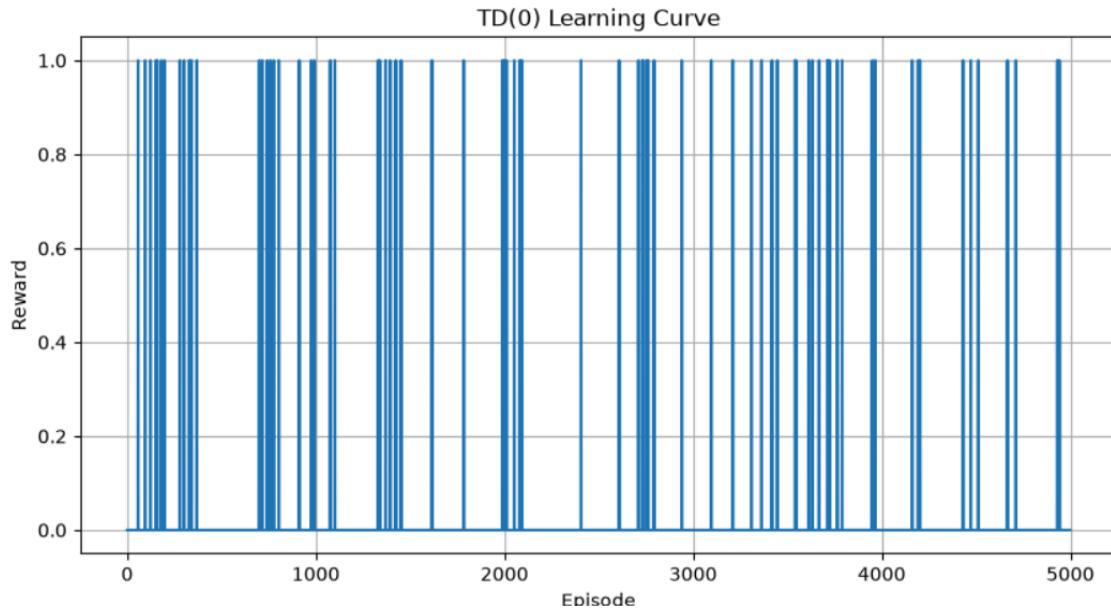
## Experiment 7 RL

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**Course Code:** MLA0305

**Slot:** B



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localhost:8888/notebooks/Experiment%207.ipynb

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JupyterLab Python 3 (ipykernel)

```
plt.figure(figsize=(10, 5))

plt.plot(episode_rewards)

plt.xlabel("Episode")
plt.ylabel("Reward")
plt.title("TD(0) Learning Curve")

plt.grid()

plt.show()

env.close()
```

=====

TD(0) LEARNING

=====

Estimated Value Function:

State 0 : 0.0038

State 1 : 0.0058

State 2 : 0.0089

State 3 : 0.0042

State 4 : 0.0031

State 5 : 0.0

State 6 : 0.019

State 7 : 0.0

State 8 : 0.008

State 9 : 0.0251

State 10 : 0.082

State 11 : 0.0

State 12 : 0.0

State 13 : 0.0871

State 14 : 0.404

State 15 : 0.0

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JupyterLab Python 3 (ipykernel)

```
# TD(0) update
V[state] = V[state] + alpha * (
    reward + gamma * V[next_state] * (not terminated)
    - V[state]
)

total_reward += reward
state = next_state

if terminated or truncated:
    break

episode_rewards.append(total_reward)

# Display Value Function
print("=====")
print("      TD(0) LEARNING")
print("=====")

print("\nEstimated Value Function:")

for state in range(n_states):
    print("State", state, ":", round(V[state], 4))

# ----- VALUE FUNCTION VISUALIZATION -----

plt.figure(figsize=(8, 5))

plt.bar(range(n_states), V)

plt.xlabel("State")
plt.ylabel("Value")
plt.title("TD(0) Estimated Value Function")
```

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JupyterLab Python 3 (ipykernel)

```
[1]: import numpy as np
import gymnasium as gym
import matplotlib.pyplot as plt

# Create environment
env = gym.make("FrozenLake-v1", is_slippery=False)

# Parameters
n_states = env.observation_space.n
alpha = 0.1
gamma = 0.9
episodes = 5000

# Initialize value function
V = np.zeros(n_states)

# Store episode rewards
episode_rewards = []

# TD(0) Learning
for episode in range(episodes):
    state, info = env.reset()
    total_reward = 0

    for step in range(100):
        # Select random action
        action = env.action_space.sample()

        # Take action
        next_state, reward, terminated, truncated, info = env.step(action)

        # TD(0) update
```

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